

Barr-coexactness of metric compact Hausdorff spaces

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110th Peripatetic Seminar on Sheaves and Logic
Tallinn, Estonia
3 May 2025

Based on:



Marco Abbadini, Dirk Hofmann.

Barr-coexactness for metric compact Hausdorff spaces.

Theory and Applications of Categories, 2025. [Link](#).

What's the opposite of

A handwritten word in cursive script, which appears to be "good". The letters are connected and fluid, with a prominent loop at the end of the word.

?

What's the opposite of

good

?

It's

good

!

How is a category with a



opposite?

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opposite?

It's



!

Various categories of “spaces” have a *good* algebraic opposite:

Category of spaces	Dual
Stone spaces	Boolean algebras [Stone, 1936]
Priestley spaces (Stone + partial order)	Bdd. distr. lattices [Priestley, 1970]
Compact Hausdorff spaces	Monadic over Set [Duskin, 1969]
Compact ordered spaces (Comp. Hausd. + partial order) [Nachbin, '40s]	Monadic over Set [A., 2019]

From order to metric

Theorem (A., 2019)

The opposite of the category of

*compact ordered spaces [Nachbin, '40s] :=
compact Hausdorff spaces + compatible *partial order**

*is monadic over **Set**.*

“*Partial order*” is similar to “*metric*” [Lawvere].

Question

Is the opposite of the category of

compact Hausdorff spaces + compatible *metric*

monadic over **Set**?

Is the **opposite** of the category of

compact Hausdorff spaces + compatible *metric*

good?

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From a different angle

Compact metric space $:=$ metric space with compact (necessarily Hausdorff) induced topology.

Example: $[a, b]^n$, as well as any closed subspace.

The category of compact metric spaces and (necessarily continuous) non-expansive maps is poorly behaved.

- Can we tweak the category of *compact metric spaces* to get a *good* category?

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- Can we tweak the category of *compact metric spaces* to get a *good* category?

Compare it with

- Is the category of “*compact Hausd. spaces + compatible metric*” *good*?

Problems of compact metric spaces

The category of compact metric spaces and non-expansive maps...

Problem 1. ... is **not cocomplete**: no coproduct of two singletons.

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Definition

A *metric* (in this talk) is a map $d: X \times X \rightarrow [0, \infty]$ satisfying:

- ▶ (symmetry) $d(x, y) = d(y, x)$;
- ▶ (reflexivity) $d(x, x) = 0$;
- ▶ (separatedness) $d(x, y) = 0$ implies $x = y$;
- ▶ (triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$.

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Problem 2. ... is **not complete**: no countable power of a two-element metric space.

Reason: the topology induced by the product metric (= sup metric) is **not** the product topology (and is not compact).

Problems of compact metric spaces

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Problem 2. ... is **not complete**: no countable power of a two-element metric space.

Reason: the topology induced by the product metric (= sup metric) is **not** the product topology (and is not compact).

Remedy: allow the topology to be just **compatible** with the metric, rather than **induced** by it.

What does “compatible” mean?

If

$$(f_n: X \rightarrow [a, b])_n$$

converges pointwise (i.e. in the product topology) to

$$f: X \rightarrow [a, b],$$

and $g: X \rightarrow [a, b]$ is another function,

$$d_\infty(f, g) \leq \liminf_{n \rightarrow \infty} d_\infty(f_n, g).$$

Definition (Hofmann, Reis, 2018)

A *metric compact Hausdorff space* is a compact Hausdorff space X equipped with a lower semicontinuous metric $X \times X \rightarrow [0, \infty]$.

Lower semicontinuous:

$$d(x_0, y_0) \leq \liminf_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} d(x, y).$$

I.e.: small topological perturbations may yield great increments in distances, but not great decrements.

Equivalently, continuous wrt the upper topology of $[0, \infty]$.

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Example: any compact metric space (in the classical sense).

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Example: An arbitrary product of compact metric spaces, with the product metric (= sup metric) and the product topology.

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Example: any compact Hausdorff space with $d(x, y) = 1$ for $x \neq y$.

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Equivalently, continuous wrt the upper topology of $[0, \infty]$.

(Replacing “metric” by “order”, we get compact ordered spaces.)

Category of metric compact Hausdorff spaces and continuous non-expansive maps =: **MetCH**.

Is **MetCH**^{op} monadic over **Set**?

I.e.:

1. Is **MetCH**^{op} complete and cocomplete?
2. Is **MetCH**^{op} Barr-exact?
3. Does **MetCH**^{op} have a regular projective regular generator?

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Is **MetCH**^{op} monadic over **Set**?

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✓ (Essentially: Tholen, 2009)
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1. Categories of “algebras”: the *good* factorization tends to be “(regular epi, mono)” $\rightsquigarrow$ regularity.
2. Categories of “spaces”: the *good* factorization tends to be “(epi, regular mono)” $\rightsquigarrow$ coregularity.

Example: **Top** is...

- ▶ ... *not* regular (proof in Borceux’ HCA2).
- ▶ ... coregular [Barr, Pedicchio, 1995].

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Example: **Top** is...

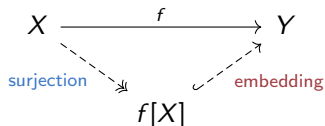
- ▶ ... *not* regular (proof in Borceux’ HCA2).
 - ▶ ... coregular [Barr, Pedicchio, 1995].
3. Categories of “(compact Hausdorff)-ish spaces” tend to be Barr-coexact.

Example:

- ▶ **Top** is *not* Barr-coexact.
- ▶ **CompHaus** is Barr-coexact [Duskin, 1969] (and [Barr & Wells, 1984] for more details).

In metric compact Hausdorff spaces

Given a morphism $f: X \rightarrow Y$ in **MetCH**, we equip $f[X]$ with the metric and topology induced by Y :



(*epi*, *regular mono*) factorization.

Theorem (A., Hofmann, 2025)

In the category **MetCH** of metric compact Hausdorff spaces,

1. *epis* = *surjective morphisms*;
2. *regular monos* = *strong monos* = *embeddings*.

Theorem (A., Hofmann, 2025)

MetCH is coregular.

$$\begin{array}{ccc} K & \hookrightarrow & X \\ f \downarrow & \lrcorner & \downarrow \\ K' & \hookrightarrow & X' \end{array}$$

Given an embedding $K \hookrightarrow X$ and a morphism $f: K \rightarrow K'$, inside X we can replace K by an exact copy of K' (and appropriate adjustments outside of K' induced by f).

Theorem (A., Hofmann, 2025)

MetCH is *Barr-coexact*.

Given a metric compact Hausdorff space X . For $K \subseteq X$ closed $\rightsquigarrow$ quotient of $X + X$ by gluing the two copies of K .

Barr-coexactness: every surjective morphism $X + X \twoheadrightarrow Z$ satisfying co-reflexivity, ~~co-symmetry~~ and co-transitivity (first-order conditions) arises in this way.

MetCH $\coloneqq$ category of metric compact Hausdorff spaces and continuous non-expansive maps. [Hofmann, Reis, 2018]

- Is **MetCH**^{op} monadic over **Set**? (I.e., equivalent to a variety of possibly infinitary algebras)? I.e.:
1. Is **MetCH** complete and cocomplete? ✓ (Tholen, 2009)
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 1. **MetCH** is not *bad*.

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- ▶ Can we tweak the category of classical “compact metric spaces” to get a *good* one?
 1. **MetCH** is not *bad*.
 - 1+2. **MetCH** is quite *good*!

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